COURSE OVERVIEW

ROPE RIGGING

The Rope Rigging and Mechanical Lifting course provides learners with the knowledge and practical skills required to plan and perform safe lifting and rigging operations in work-at-height environments. The course focuses on correct equipment selection, inspection, configuration, and controlled lifting practices.

Learners are introduced to load calculations, anchor selection, mechanical advantage systems, centre of gravity, and safe lifting techniques. Emphasis is placed on risk awareness, correct system setup, communication during lifting operations, and compliance with site safety requirements.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

- Identify hazards and risks associated with rigging and lifting operations
- Select, inspect, and use rope rigging and mechanical lifting equipment
- Explain load ratings, safety factors, and basic load calculations
- Identify and use suitable anchor points for rigging systems
- Apply basic mechanical advantage systems for lifting and lowering
- Contribute to safe lifting operations through communication and planning

INDEX

INTRODUCTION

GRAVITY VALUES

R — RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.

I — INTEGRITY

Conducting day-to-day business with integrity at all times.

S — SOLUTIONS DRIVEN

Being solution driven.

Q — QUALITY & EXCELLENCE

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

SITE ESTABLISHMENT

Before any work begins, the site must be safe.

Use this simple checklist

Follow the Rules:
Make sure you follow all client rules.

Example: If night work is not allowed, do not work at night.

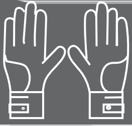
01

Rescue Plan
Have a rescue plan ready. A rescue kit and first aid kit must be on site.

02

Work in Pairs
There must be at least two workers on site.
If something goes wrong, one can help or call for help.

03



Check Equipment
All tools and gear must be safe and working before use.

04

05

Safe Site
Check the building or structure.

Make sure it is strong and safe to work on.

06

Manage Traffic

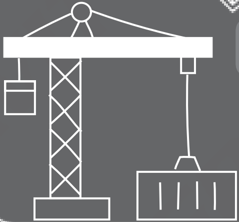
If there are cars or trucks nearby, there must be a traffic safety plan.

07

Do a Risk Assessment

Check what dangers there are.

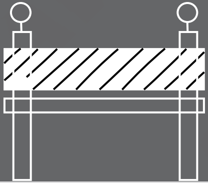
All workers must understand the risks before starting work.



08

Use the Right Safety Gear

Make sure the fall protection fits the job and the site.



EXCLUSION ZONES **& BARRICADES**

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

SITE SAFETY

Be healthy and ready before starting work.

Only do jobs you are trained and allowed to do.

Never work under the influence of alcohol or drugs.

Always wear your safety gear (PPE, harness if working at height).

Use harnesses correctly and stay connected at heights.

Never work alone and always follow safety steps.

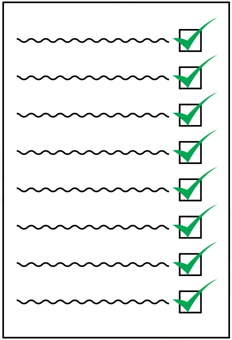
GRAVITY

SITE SAFETY

- **Keep people away** from areas below when working above.
- **Don't walk under lifted loads** or stand in dangerous areas.
- **Only trained people** can do electrical, underground, or confined space work.
- **Follow road rules.** Wear helmets. No passengers on open trucks..
- **Don't drive tired** or distracted.
- **Stop unsafe work!** Everyone has the right to speak up.

A. Administrative Requirements for Lifting:

Before commencing a lift, the following practices must be followed.



1. **Lifting Plan:**
   A risk assessment for rigging procedures.

2. **Risk Assessment:**
   A documented control of risks on site.

3. **Gear Inspection and Records:**
   Pre-use and long-term inspection records

4. **Pre-Lift**:
   A 5-10cm test-lift to ensure that all is well and that it is safe to proceed.

Detailed lifting plan:

Site details:

Site Name : ____________________ *Site No :* ____________________

Site Supervisor Name : ____________________

Contact Number : ____________________ *Date :* ____________________

No. of workers on site : ____________________

Lift details : ____________________

____

____

Type of anchor pionts with standard and rating:

____

Methods of communication during the lift : HAND SIGNALS

TWO WAY RADIO ☐ VERBAL ☐

Admin requirements : *(These requirements are the responsibility of the Site Supervisor as mentioned avove. e.g.Training, PPE, Inspection etc)*

1. ________________________ 2. ________________________

3. ________________________ 4. ________________________

5. ________________________ 6. ________________________

Equipment requirements : *(Include SWL or WLL per item)*

Rope rigging equipment	Qty	Mechanical lifting equipment	Qty

Lift 1 load discription :

Weight of the load in Kg : ____________________ *Shape of the load :* ____________________

Pulley System to be used : ____________________ *Estimate tag line weight* ____________________

All up weight : ____________________ *Top anchor load in Kg :* ____________________

Height of the lift in meters : ____________________ *Lift* ☐ *or Lift & lower* ☐

Team compesition for lift 1: ____________________

Draw a detailed plan with the type of system that will be used for the lifting of the load. Indicate the weight of the load and the weight that will be placed on the top and bottom anchor once the lift starts while considering the weight that the tagline forces will add.

Lift 1 sign off:

Lift planner and supervisor: I confirm that I have planned this lift in accordance with IBP procedures and accept the responsibilities of my position.

Initials and Surname	Signature	Date

Change notes:

(Should anything happen during the lift causing a deviation from the lifting plan, please note it here, this includes: the use of lifting equipment and relevant accessories)

Change notes: sign off:

Lift planner and supervisor: I confirm that I have planned this lift in accordance with IBP procedures and accept the responsibilities of my position.

Initials and Surname	Signature	Date

Pre-operational checks:

- [ ] NO VEHICLES ARE TO BE USED FOR LIFTING
- [ ] THE EFFECT THAT THE WIND HAS ON THE LOAD HAS BEEN CONSIDERED
- [ ] THE EFFECT OF THE TAG LINES HAS ON THE LOAD HAS BEEN CONSIDERED
- [ ] ALL FORCE WILL BE APPLIED AT BASE OF TOWER SHOULD THE HAULING LINE BE DEVIATED
- [ ] NO ONE TO STAND UNDERNEATH LOAD
- [ ] NO BRAIDED ROPES ARE TO BE USED
- [ ] NO NYLON (SKI) ROPES ARE TO BE USED
- [ ] RIG FROM ANCHORS OF SUITABLE STRENGTH (NO FENCES, VEHICLES ETC)
- [ ] THE CENTER OF GRAVITY OF THE LOAD HAS BEEN CONSIDERED
- [ ] THE LOAD HAS BEEN CORRECTLY SLUNG
- [ ] ONLY EQUIPMENT THAT IS SAFE FOR USE AND FIT FOR PURPOSE IS TO BE USED
- [ ] THE LIFTING/LOWERING ROUTE IS CLEAR OF OBSTRUCTIONS
- [ ] THE LOAD DOES NOT EXCEED WLL OF EQUIPMENT
- [ ] THE BANKSMAN IS VISIBLE AT ALL TIMES
- [ ] WEATHER CONDITIONS HAVE BEEN CONSIDERED
- [ ] NO WORKER TO CLIMB WITH LOAD (excl. tools) CONNECTED TO THEMSELVES
- [ ] NO UNDERGROUND SERVICES, OVERHEAD POWERLINES ETC
- [ ] STABILITY OF STRUCTURE AND LIFTING AREA CONSIDERED
- [ ] ALL WORKERS ON SITE HAVE RELEVANT COMPETENCIES
- [ ] RELEVANT PERMISSIONS HAVE BEEN OBTAINED

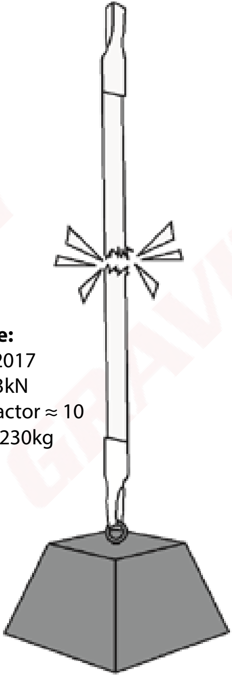
Agree and confirm:

I, the undersigned, agree that the I understand and accept the plan as written above.

Name and Surname	Designation	Signature
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
11.		
12.		
13.		
14.		
15.		

B. Rope rigging equipment

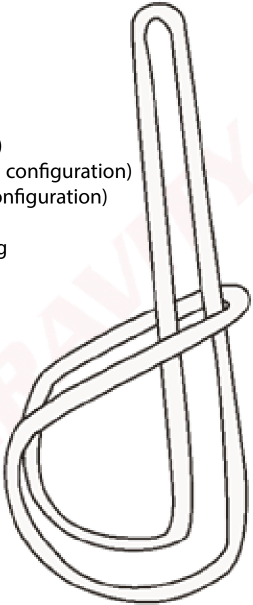
1. Breaking Strength (B.S.): refers to the amount of tensile force needed to break the piece of equipment. With regards to work at height PPE, manufacturers will always provide this amount of force in Kilonewtons (kN). In order to simplify calculations an approximate conversion from force (vector) to weight (scalar) is made in the form of 1kN ≈ 100kg. The Breaking Strength does change and depends on the configuration in which the equipment is used.



Example:
EN 566:2017
BS >≈ 23kN
Safety Factor ≈ 10
SWL <≈ 230kg

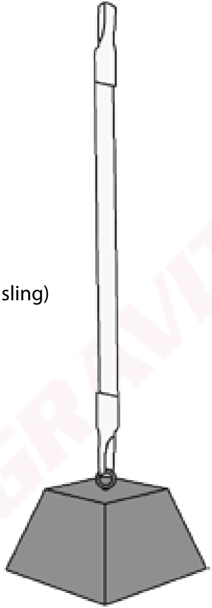
2. Safe Working Load (S.W.L.): refers to the safe amount of weight that is allowed to be loaded on a piece of equipment and must be calculated by the user. The SWL is equal to the BS in (kN) divided by a safety factor of 10. The safety factor is employed to ensure, that should a weight fall while connected to the piece of equipment, the resultant force due to acceleration is still below the BS. Thus a SWL needs to be calculated and converted from kN to kg, SWL (in kg) ≈ (BS /10)*100. The SWL depends on the configuration in which the equipment is used. See example: The sling configuration is changed due to choking the sling which in return reduces the sling SWL by 20%.

Example:
EN 566:2017 (Choked)
BS >≈ 23kN (Standard configuration)
BS >≈ 17kN (Choke configuration)
Safety Factor = 10
SWL< = 1.7kN = 170kg



3. Working Load Limit (W.L.L.): With regards to lifting tackle, the manufacturer will always provide the Working Load Limit which indicates the maximum amount of weight that may be lifted with the equipment. This will be based on a pre-determined safety factor. This will be given either in kg or Tons. The WLL depends on the configuration in which the equipment is used.

Example:
EN 1492-1:2000
WLL = 1T (purple sling)
Safety Factor = 7
M.B.S = 7T



4. Safety Factors

DIFFERENCES BETWEEN LIFTING EQUIPMENT, AND ROPE ACCESS & FALL ARREST EQUIPMENT:

Lifting Equipment & Safety Factors:

- Equipment used primarily for lifting operations.
- Testing certificates issued annually.
- Lifting equipment will always have a W.L.L. (Working Load Limit) that indicates the maximum load that can be lifted with the equipment.
- The rating will be in TON or KG.
- Lifting equipment also has a safety rating, for example 6:1, which implies that a 1 ton webbing sling will break at 6 ton. This will be indicated on the equipment.
- Lifting slings are colour coded according to strength and should not be confused with Rope Access or Fall Arrest equipment which is not colour coded.

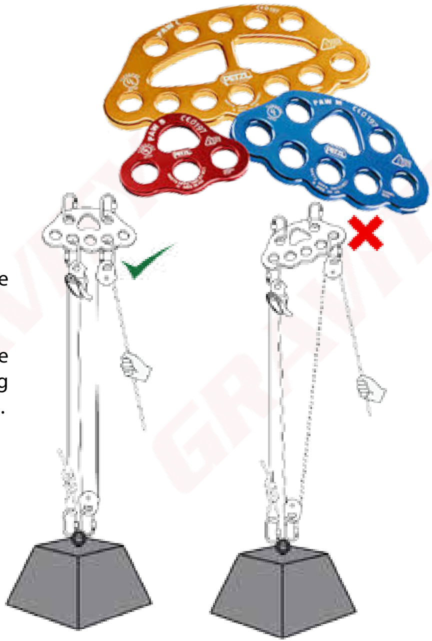
Rope Access & Fall Arrest Equipment, and Safety Factors:

- Equipment for the work at height industry. Can be used to support equipment and people.
- Equipment inspections done at least every 6 months by a competent equipment controller.
- Rope Access and Fall Arrest equipment will have a S.W.L (Safe Working Load) on the equipment that will indicate the amount of weight (in KN; 1 KN = 100 kg) that can safely be applied onto the equipment.
- However, the M.B.S. (Minimum Breaking Strain) of equipment indicates the amount of weight the equipment can hold before it will break.
- In Rope Access and Fall arrest equipment the relation between the S.W.L. and the M.B.S. will be the safety rating (not indicated on the equipment) and will always be 10:1.

5. Range of Equipment

Rigging Plates (Petzl Paw)

- For easy organization of the work area and for creating multiple anchor systems.
- Must be loaded evenly.



A Note On Rigging Plates

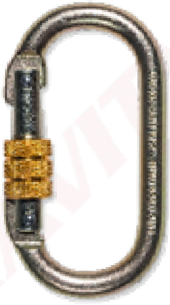
When using rigging plates, always work from the inside outward and not form the outside inwards.

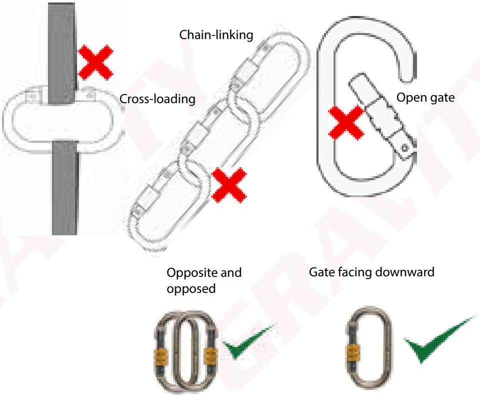
Working from the inside outward will cause the rigging plate to be better balanced, thus ensuring the load is split equally over multiple anchor points.

- M.B.S. :3600 kg
- S.W.L. :360 kg

Karabiners (EN 362)

Used to connect a range of devices and different pieces of equipment.





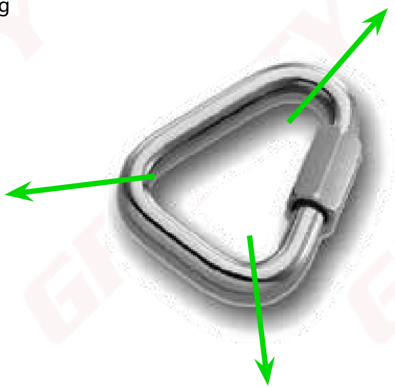
- Always lock karabiners.
- Ensure that the screw gate closes down.
- Always load a karabiner on the spine.
- Do not chain-link karabiners.
- Do not cross-load.
- Ensure gates are opposite and opposed.

- M.B.S. :2200 kg
- S.W.L. :220 kg

Delta Maillon (EN 362)

Used for connecting equipment if no rigging plate is available, because:
Delta Maillons can be loaded in 3 directions.
Similar to a karabiner, but is most commonly used for more permanent connections.

- M.B.S. :2200 kg
- S.W.L. :220 kg



Singing Rock Extra Roll Pulley (EN12278)

- Used for lowering and lifting.
- Designed to manoeuvre heavy loads or for intensive use.
- Large diameter sheave mounted on sealed ball bearings for efficiency (bigger pulley = less friction).
- Accommodates up to three karabiners.
- Both sides must be clipped into karabiner/s.
- Only use "oval" type karabiners.
    - M.B.S. :3200 kg
    - S.W.L. :160 kg

Camp Tethys Pro (EN12278)

- Used for lowering and lifting.
- Sealed ball bearing for efficiency
- Both sides must be clipped into karabiner.
- Only use "oval" type karabiners.
- Due to smaller sheave there is more friction.
    - M.B.S :2500 kg
    - S.W.L :300 kg

Progress capture pulley (EN 12278 & EN 567)

- A highly effective progress capturing device for one-way lifting operations.
- Only use "oval" type karabiners.
  - S.W.L. :320 kg
  - M.B.S. :3200 kg
  - S.W.L .(with cam engaged) :100 kg

Descender (EN341, 12841 type C)

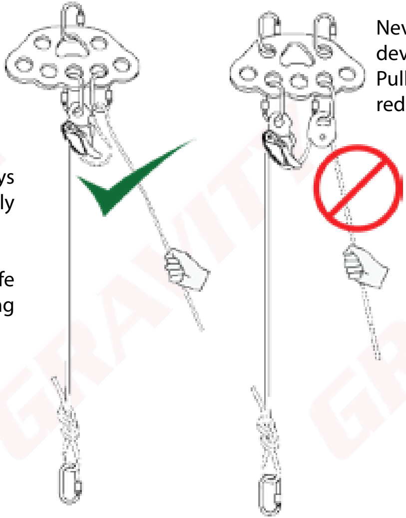
A variable friction device that controls the rate of descent. Used for an emergency escape, horizontal lifelines , rescue and rigging operations.

- When in use during lower operations, extra friction must be applied.

- S.W.L. = 120kg
- M.B.S. = 1200kg
- S.W.L (with extra friction) = 220kg

When lowering a load, always ensure extra friction is correctly applied.

Extra friction also increases the safe working load of certain descending devices.



Never use a pulley as a friction device.
Pulleys are made and designed to reduce friction.

Rope Grab (EN353-2, 358)

This device is a rope grab designed to stop a rope from running out of control, thus it is used in a lifting system to back up the main hauling system.

- S.W.L :150 kg
- M.B.S :1500 kg

Rope Clamp Devices

Camp Turbohand (EN567)

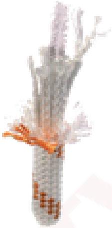
- Designed for ascending and occasionally for hauling as it bites into a rope, stopping it from sliding DOWN, but can easily move UP a rope.
- The Ascension is used with an Etrier to help with lifting.
- Never use an EN567 device as a backup device as it is an aggressive device and can cause rope damage.
    - Rope Damage :400 kg
    - M.B.S :1000 kg
    - S.W.L :100 kg

Camp Turbochest (EN567)

- Same as the Ascension but is used to rig a 3:1 system on a rope.
- Never use an EN567 device as a backup device as it is an aggressive rope device.
    - Rope Damage :400 kg
    - M.B.S :1000 kg
    - S.W.L :100 kg

Ropes

Semi-Static (EN1891)



- Must be used in a rigid system (a system where there is no chance of shock load).
- Low stretch: 5% to 10%
- 8 - 13 mm in diameter.
- Kernmantle construction: Made of nylon or polyester.
  Mantel (sheath) protects the kern (core).
- Mantle (20%), Core (80%).

Dynamic (EN892)

- Used for dynamic systems where ropes serve as shock absorber.
- High stretch: between 10% and 20%
- Used only for tag lines in rigging.
- 10.5 mm to 13 mm in diameter.
- Kernmantle Construction: Made of nylon or polyester. Mantle (sheath) protects the kern (core).
- Mantle (20%), Core (80%).
  - S.W.L :250 kg (10.5 mm)
  - S.W.L :300 kg (11.5 mm)
  - S.W.L :400 kg (380 kg with Fig 8 knot) (12.5 mm)

NOTE: For safe and effective rigging we will be using semi-static ropes as we are trying to keep our rigging system as rigid as possible.

Rope Care

- Do not use rope for anything other than work at height.
- Do not run rope over sharp edges; use edge protection.
- Do not stand on or walk over a rope.
- Ensure that all devices are compatible with any devices to be used.

Rope Bagging:

It may be preferable to bag a rope instead of coiling it. In such cases, it is recommended to deviate the rope over the structure or ladder and pull the rope into the bag, this will prevent knotting and allows a rope to be bagged faster.

NOTE: Remove ropes from service as soon as they are damaged, e.g. frayed, burned or cut.

Edge protection

- Edge protection is needed where the possibility of rope damage can occur due to sharp edges.
- All edges with a diameter of less than 5 mm is considered a sharp edge and edge protections should be used.

Types of Edge Protection

- Rollers: Placed on the sharp edge, the rollers also minimize rope drag when retrieving your rope.
- PVC edge protectors: can also be used, as well as rope mats.

Anchor Sling (EN566)

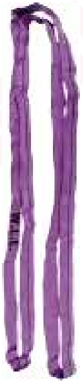
- Both D-links must be connected via karabiner. (May have a PVC protective cover)
- Used for anchoring (wrapped around a self-identified anchor point).
- Must be stitched, not tied.
  - M.B.S :2200 kg
  - S.W.L :220 kg

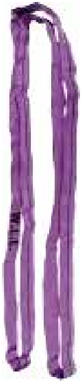
Open Round Sling (EN566)

- Both ends must be clipped with a karabiner.
- Used for anchoring (wrapped around a self-identified anchor point).
- Must be stitched, not tied.
- NOT colour coded.
  - M.B.S :2200 kg
  - S.W.L :220 kg

Webbing Lifting Slings (EN 1492-1:2008):

- Webbing Lifting Slings are color coded per lifting rating: e.g. green is for 2 ton WLL etc.
- Webbing slings have a safety rating of 6:1 (unless otherwise stated by the manufacturer) which means that a 1 T sling will break at 6 T.
- Colour coded as per EN 1492-1.
- Must be inspected annually by an lifting machinery inspector (LMI).
- User must conduct pre-use inspections as per software requirements.



		Working Load Limits with 1 Webbing Sling						
	Colour coded according to DIN-EN 1492-1	Straight Lift	Choked Lift	Basket 0 Degrees	Basket 0-7 Degrees	Basket 7-45 Degrees	Basket 45-60 Degrees	Basket 45-60 Degrees
								
		1.0	0.8	2.0	1.4	1.0	0.7	0.5
	WLL 1T	1000	800	2000	1400	1000	700	500
	WLL 2T	2000	1600	4000	2800	2000	1400	1000
	WLL 3T	3000	2400	6000	4200	3000	2100	1500
	WLL 4T	4000	3200	8000	5600	4000	2800	2000
	WLL 5T	5000	4000	10000	7000	5000	3500	2500
	WLL 6T	6000	4800	12000	8400	6000	4200	3000
	WLL 8T	8000	6400	16000	11200	8000	5600	4000
	WLL 10T	10000	8000	20000	14000	10000	7000	5000
	WLL 12T	12000	9600	24000	16800	12000	8400	6000

Note: When changing the configuration of the slings the WLL will either increase or decrease the strength of the sling.

Polyester Rope Sling (EN 1141)

- Both ends must be clipped with a karabiner.
- Used for anchoring (wrapped around a self-identified anchor point).
- Must be stitched, not tied.
  - M.B.S :2200 kg
  - S.W.L :220 kg

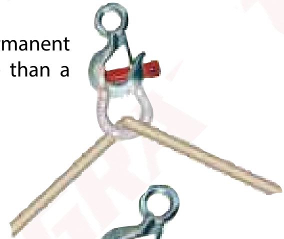
Wire Rope Slings (SANS 7531)

- Wire rope slings are used where sharp edges may damage webbing or rope slings.
- Wire rope slings are very strong and can range from a breaking strength from 45kN to 4000kN
- Both ends must be clipped with a karabiner.
- Must be inspected annually by an lifting machinery inspector (LMI).
- User must conduct pre-use inspections as per hardware requirements.

Shackles (EN 13889)

- Bow shackles are available from 0.33 Ton up to 120 Ton.
- The most common shackle is the Screw Pin Bow Shackle.
- Must be inspected annually by an lifting machinery inspector (LMI).
- User must conduct pre-use inspections as per hardware requirements.

Advantages of Bow Shackles:



1. The advantage of a bow shackle above the D-shackle is the fact that due to its shape it can accommodate more attachments.
2. In environments where vibration is a problem or more permanent attachment is needed, the Bow Shackle is a better choice than a Karabiner or Maillon.

Safe Practice (uses) of Bow Shackles:

1. Ensure that ONLY rated shackles are used and not shackles that are available from retail shops and that are used for decoration etc.
2. Ensure that with Bow Shackles with safety pins, the split pin is inserted and properly split and bent open to ensure no accidental undoing of the nut is possible.

Chain Slings:

- Chain links are available from 1 Ton up to 100 Ton.
- Chain links also vary in number of legs which affects the W.L.L.
- Must be inspected annually by an lifting machinery inspector (LMI).
- User must conduct pre-use inspections as per hardware requirements.

Use of chain slings:

1. Never make knots in the chain.
2. Never exceed 120° between the chain legs.
3. Make sure that the load is evenly distributed between all legs.
4. All legs not in use should be hooked back on the oval to avoid swinging or snagging.
5. Always refer to manufacturer's recommendations for safe use instructions

Eyebolts

- Eyebolts are available from 0.07 Ton up to 8.6 Ton.
- Eyebolts are only to be connected to points that are specified by the manufacturer.
- Must be inspected annually by an lifting machinery inspector (LMI).
- User must conduct pre-use inspections as per hardware requirements.

Use of eyebolts:

1. Store and handle eye bolts correctly.
2. Inspect eye bolts before and after use.
3. Ensure the eyebolt is suitable for the task at hand.
4. Ensure that the eye bolt and connection points are compatible and strong enough for the imposed load.
5. Ensure that the collar is fully seated when hand tight.
6. Always refer to manufacturer's recommendations for safe use instructions

Plate Grabs

- W.L.L. varies from 1ton to 10ton.
- Used to clamp onto sheets or plates in order to lift them more easily.
- Different plate grabs are available for horizontal, vertical or universal lifting operations.
- Must be inspected annually by an lifting machinery inspector (LMI).
- User must conduct pre-use inspections as per hardware requirements.

Use of plate grabs:

1. Ensure the clamp is suitable for task at hand
2. To not allow load to jerk.
3. Always adhere to manufacturer's W.L.L. restrictions
4. When loading always use the whole of the jaw.
5. Never lift more than one plate at the a time.
6. Never weld anything to a plate grab.
7. Always refer to manufacturer's recommendations for safe use instructions

Pre-use Inspections

Users must inspect all the equipment to be used before work starts. This must be done to ensure that all equipment is safe for use and fit for purpose in accordance with ISO 22846-2.

Equipment is divided into 2 categories namely hardware and software, each of these have their own inspection criteria. Some pieces of equipment (e.g a harness) fall into both categories and must be inspected as such.

Software

Check for:

- Date of manufacture
- Condition of stitching
- Discolouration
- Cuts
- Burns
- Wear and tear
- Certification

Hardware

Check for:

- Rust
- Working action
- Movement
- Wear and tear
- Deformation
- Cracks

Housekeeping:

Whilst on site, it is important to ensure that the equipment being used is handled in such a way that they do not suffer any unnecessary damage or exposure to environmental conditions. When equipment is not in use , store them in their bags or bins. Ensure that equipment is moved out of the areas where workers move to prevent slips and trips or other accidents.

Storage:

- Gear must be stored in a cool, dry environment.
- Gear must be stored in allocated places according to work site procedures.
- Gear can safely be transported using PVC covered gear bags.
- Store in designated space or dedicated rope bag.

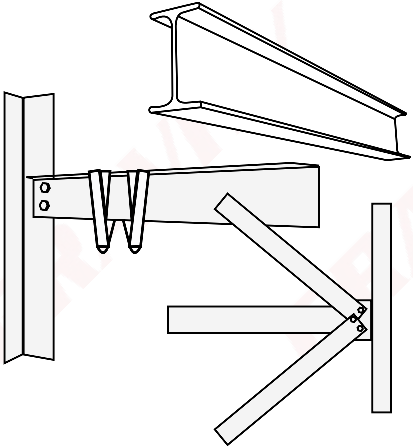
Lifespan:

Confirm the lifespan of equipment with the manufactures' specifications. Lifetime is generally 10 years, unless otherwise specified by the manufacturer, from the date of production with the exception of metal components (connectors, braking devices, etc.) for which the lifetime is determined only by frequency of the use, wear and tear, and working action.

C. Anchor Points

Anchor points are mainly divided into two categories:

- Self-identified anchor points, and
- Planned anchor points (EN 795:2012).



Gravity Access Anchor

Eye-nut

Self-identified

- S.W.L. = 150kg
- M.B.S. = 1500kg

Planned

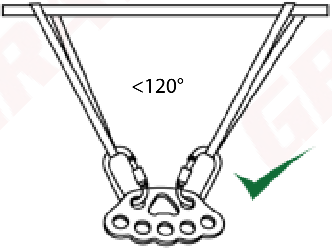
- S.W.L. = 120kg
- M.B.S. = 1200kg

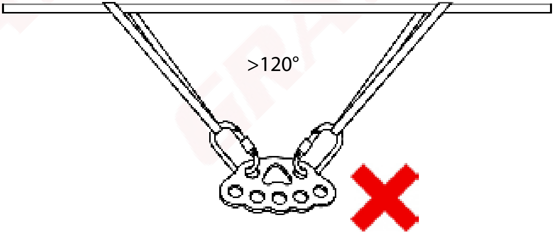
1. Anchor Points for Rigging

- An anchor point is a fixed object to which ropes or equipment may be attached.
- Anchor points should be strong enough to withstand the forces generated by the rigging system.
- Use two separate anchor points where possible or when in doubt.

2. Load Sharing Anchors (LSA)

As far as possible, it is always a good idea to share the load between two anchor points. However, it is important to remember that angles play a big part when sharing loads, and angles should be kept to a MINIMUM.





120° MAX

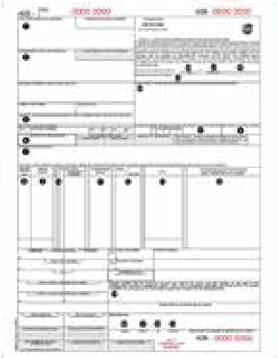
D. Lifting

Load Calculations

Proper rope rigging reduces the risk of injury and the damaging of equipment. In order to determine the correct rigging procedure and equipment for the job at hand, an accurate load calculation will need to be conducted to ensure that the load required to be lifted is within the W.L.L. of the rigging equipment.

Two ways are generally used to determine the weight of a load:

- Load cell
- Waybill

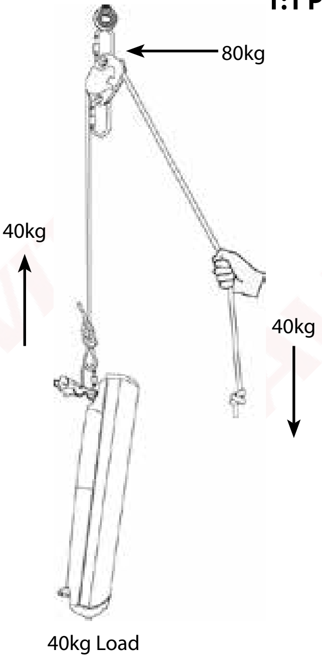


1:1 Pulley System

1. Mechanical Advantage Systems

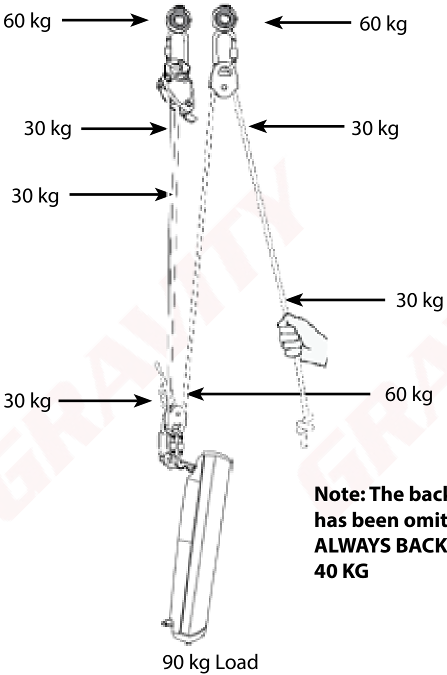
There are various mechanical advantage systems or pulley systems. All of them have their own advantages and disadvantages. Here are a few examples

When using a 1:1 Pulley system (as shown to the right) for a load between 1 and 40 kg, the typical team composition will consist of 2 persons, 1 supervisor and 1 hauler.



3:1 Pulley System

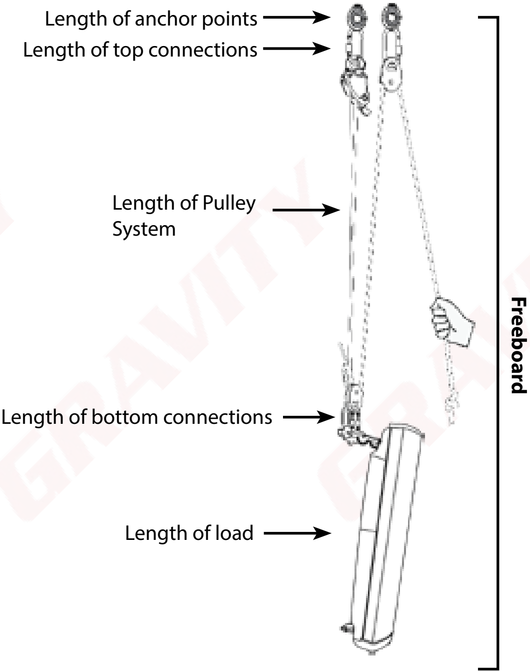
When using a 3:1 Pulley system (as shown to the right) for a load between 40 and 100 kg, the typical team composition will consist of 3 persons, 1 supervisor and 2 haulers.



Note: The back-up system has been omitted for clarity, ALWAYS BACK UP LOADS > 40 KG

Freeboard

Freeboard refers to the minimum amount of space a rigger would need to lift a load to the desired height. The freeboard calculations are made as follows:



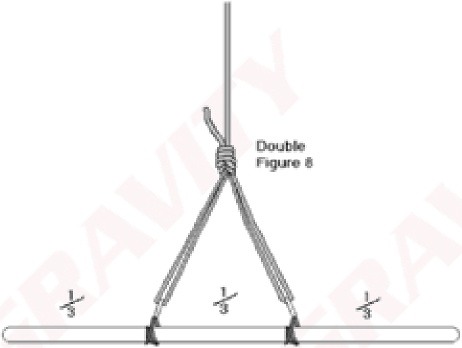
Slinging Principles

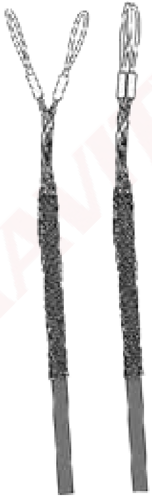
1. Ensure that the slinging is done in such a way that the load is as secure in the air as it would be on the ground.
2. The slinging method should be adequate to the type of load, with attachment point to the load and the equipment.
3. The weight of the load should not exceed the SWL of the slinging gear.
4. The load must not damage the slinging gear and the slinging gear must not damage the load.
5. Know the weight of your load.
6. Consider the centre of gravity of the load.
7. Two competent slingers/ riggers must be on site, one at the bottom and one on the structure.

Centre of Gravity:

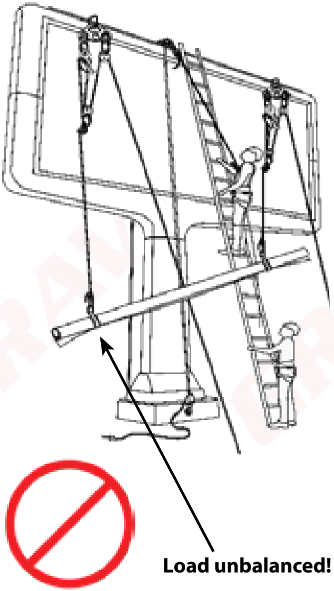
- Very careful consideration has to be given to the centre of gravity of a load. This has to be considered on the horizontal and vertical axis.
- The sling lengths might have to be adjusted so that the centre of gravity is under the hook with the load level.
- When this adjustment is carried out, calculations will be required.

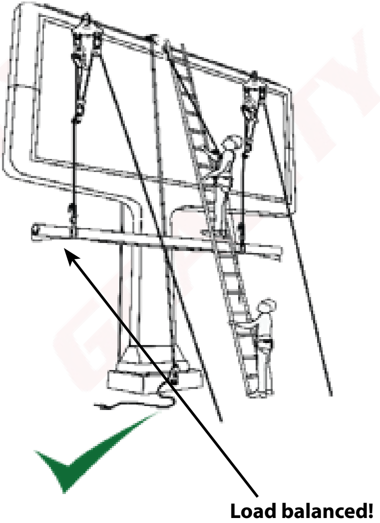
Examples of slinging methods:
- Horizontal orientation, and
- Vertical orientation.





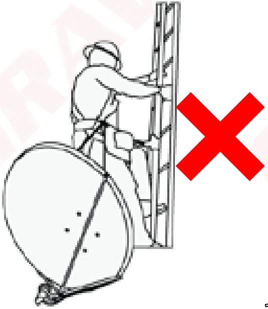
Ensure that, when lifting, the load remains stable and even



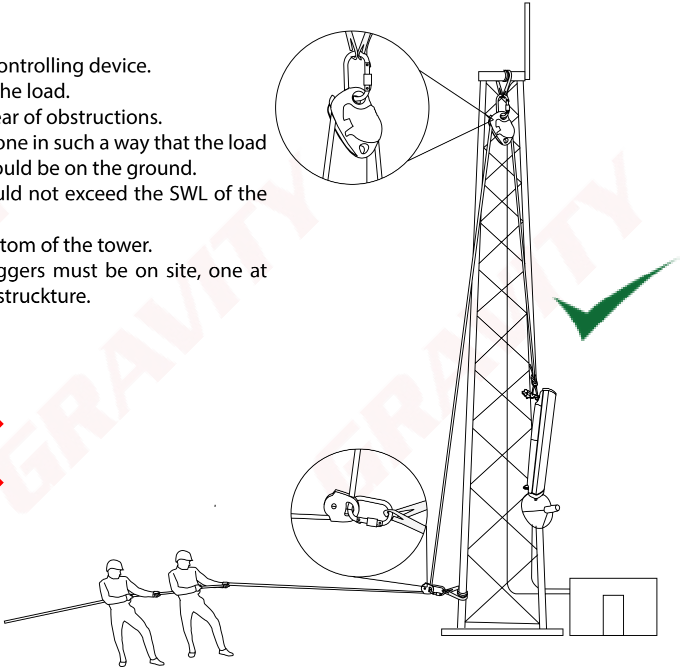


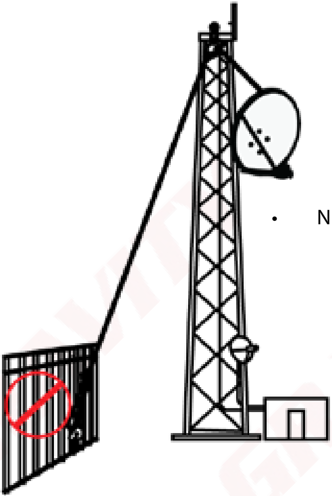
Rigging Principles

1. Always be in reach of your controlling device.
2. Never stand directly below the load.
3. Ensure the lifting route is clear of obstructions.
4. Ensure that the slinging is done in such a way that the load is as secure in the air as it would be on the ground.
5. The weight of the load should not exceed the SWL of the slinging gear.
6. Always deviate from the bottom of the tower.
7. Two competent slingers/ riggers must be on site, one at the bottom and one on the struckture.



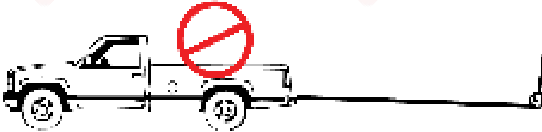
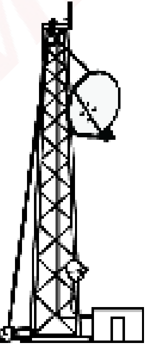
NEVER carry dishes up!



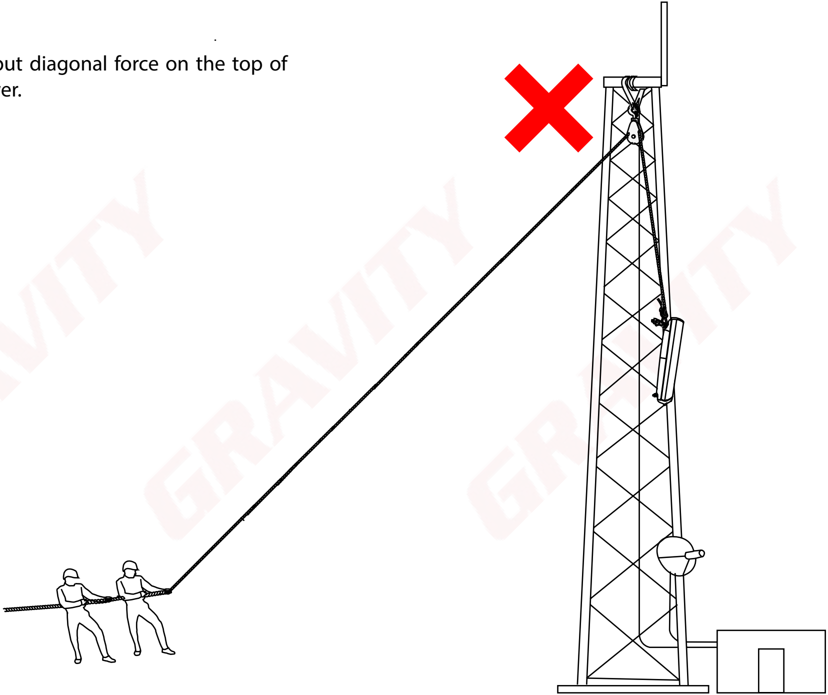


- NEVER rig of fencing!

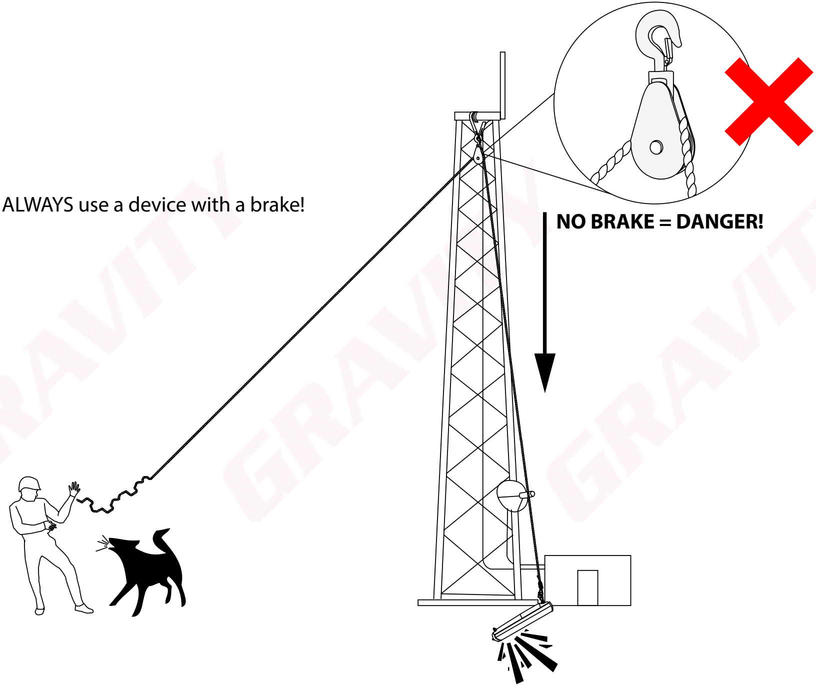
- NEVER use vehicles for lifting!

- Never put diagonal force on the top of the tower.

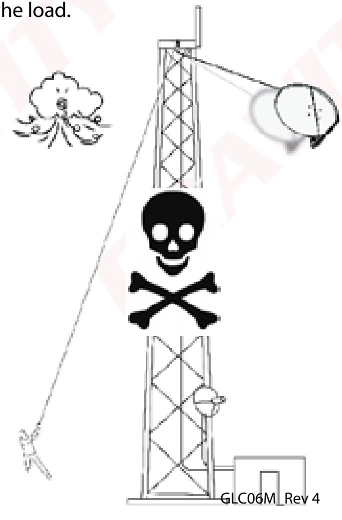


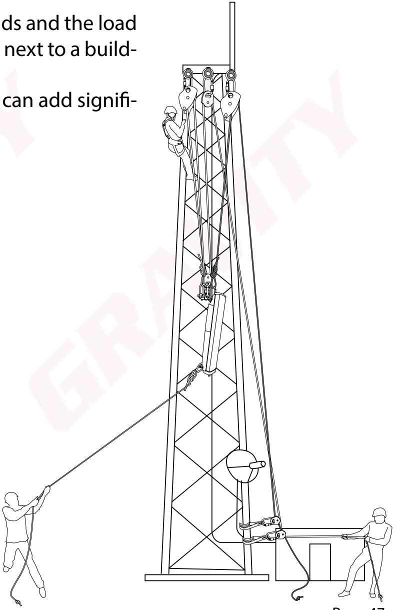
- ALWAYS use a device with a brake!



Taglines:

- Taglines (Guide Ropes) are ropes attached to an object to steady, guide or secure the object.
- Double guide ropes are used where there might be high winds and the load needs to be kept steady or if a lifting operation that happens next to a building with glass panes or protruding edges or antennas.
- Careful calculations must be made when using taglines as it can add significant weight to the load.





Hand Signals Used During Lifting

Hand signals are an easy, effective and inexpensive way of communication. Because they are visual, they are almost unmistakable if kept simple. They are also universal, thus language barriers are eliminated.

Some guidelines to follow regarding the use of hand signals:

- All lifts should employ a signalman using standard hand signals.
- All signalmen must be qualified.
- The signalman must always be in clear view of the crane operator.
- Signalman must always maintain visual contact of the load and the hoisting equipment.
- The signalman must act as the safety watch to keep the lift site free of unauthorized personnel.
- There should be only one designated signalman for each lift.
- The signaller should wear distinctive clothing.

BASIC HAND SIGNALS

Lift Load

Lower Load

Lower Load Slowly (Pre-Lift)

Lift Load Slowly (Pre-Lift)

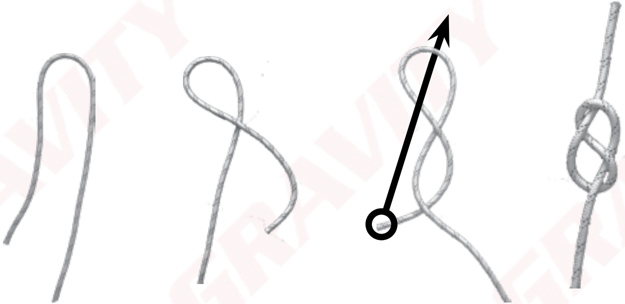
Stop

Dog Everything
(Pause until further notice)

Knots

- All knots weaken and decrease the length of the rope.
- All knots must be dressed properly; an incorrectly dressed knot can be up to 50% weaker than a knot dressed correctly.

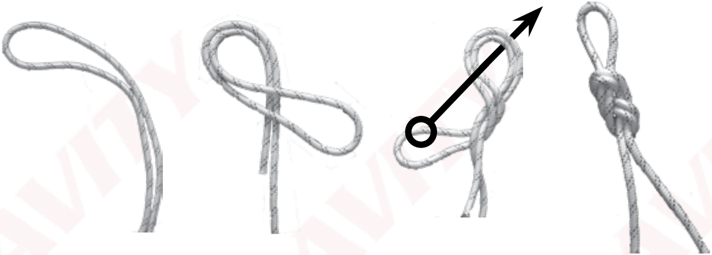
1. Figure 8 Stopper Knot



This knot is made in the end of a rope to stop people and equipment from being able of move off the end of the rope.

The tail must be longer than your fist, but shorter than your forearm.

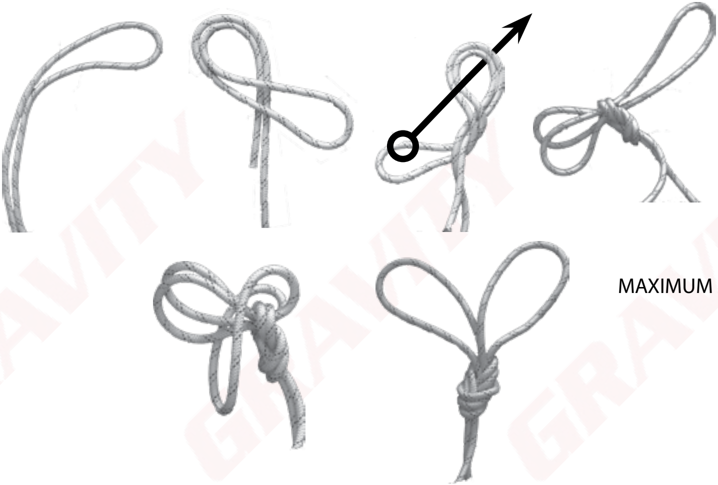
2. Figure of 8 on a Bight



This knot is used to connect the rope to a single anchor point or device. much easier to untie than the overhand, it does not have the same tendency to jam and so injure the fiber, and is larger, stronger, and equally secure.

Figure 8 on a Bight = 80% of rope strength

3. Double Loop Figure 8



MAXIMUM ANGLE = 90°

This knot is easier to untie than a plain Figure Eight on a Bight and is used to connect a rope to two separate anchor points and is able too adjust to compensate for center of gravity balancing.

Figure 9 on a Bight = 80% of rope strength

4. Figure 9 on a Bight

This knot is used for similar reasons as the Figure 8 on a Bight, however the Figure 9 is better suited when lifting heavier objects as it avoids sharp bends and, therefore, the rope retains about 80% of its strength and it unties more easily after it has been loaded.

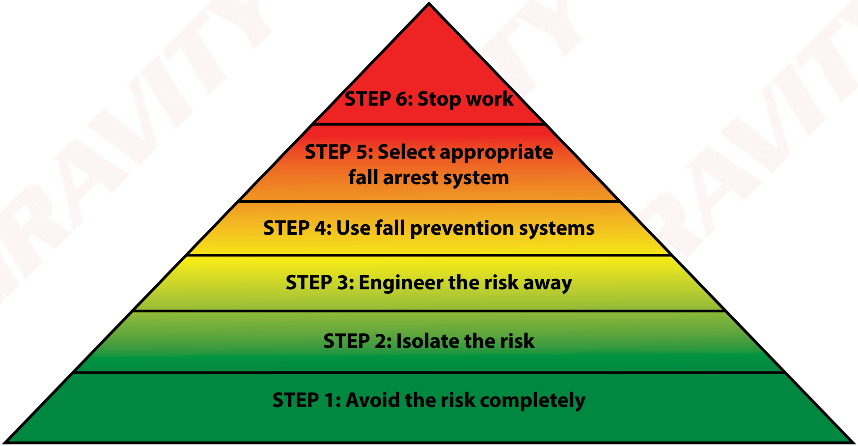
5. Klemheist

Used to create an anchor point on a load which needs to be lifted vertically. It is recommended to use the klemheist in conjunction with half hitches. It is easy to tie, and is reliably secure when the loop is loaded and can be adjusted when the loop isn't loaded

E. Risk Assessment

A risk assessment must be conducted by the site supervisor on site, daily, before work starts in order to ensure that all risks on site are controlled and mitigated.

Below is a flowchart depicting the process flow for risk mitigation



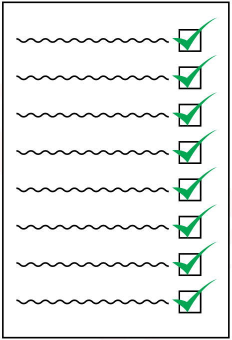
- **Adverse Weather Conditions:**
  No rigging work-at-height shall be done during any inclement weather conditions. Should the situation change whilst at height, all technicians must safely descend to the ground and the Site Supervisor must report back immediately to the Project Manager.

  Examples of inclement weather conditions:
  - Rain
  - Lightning
  - Extremely cold weather
  - Extremely hot weather
  - Sleet or snow
  - High or gusty wind (keep in mind shape and weight of load)
  - Poor visibility (fog or smog)

F. Lifting Plan

A lifting plan is a document specifying the how and by whom the lift will be completed. This must be completed by the Site Supervisor on site before work starts, and must include the following:

- The site name/address
- An approved work permit
- The system to be used (with back-up, taglines etc)
- The team composition
- Description of the lift
- The type and weight of the load
- Drop zones and exclusion zones
- Anchor points to be used
- Weather conditions
- The expected duration of the lift
- The supervisor's details
- Client details



If at any point alterations have to be made to the lifting operation which were not indicated on the lifting plan, it must be documented and signed by the supervisor before continuing with the lift.

Congratulations on completing Gravity Training's Rope Rigging course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map or consult your facilitator/assessor.